

Kiran VADDI

 kiranvad.github.io  kiran.vaddi@nist.gov
 [linkedin.com/in/kiranvaddi27](https://www.linkedin.com/in/kiranvaddi27)  github.com/kiranvad  Google Scholar
 100 Bureau Dr, Gaithersburg, MD 20899

EDUCATION

June 2021 May 2025	University of Washington, SEATTLE, WA, USA Post Doctoral Training in Chemical Engineering Functional Space Data Representations for Autonomously Engineering Soft Materials
May 2021 August 2017	University at Buffalo, The State University of New York, BUFFALO, NY, USA Ph.D. in Materials Science and Engineering Thesis : Representations for data-driven material discovery
June 2016 June 2012	Indian Institute of Technology Madras, CHENNAI, TN, India Dual Degree in Mechanical Engineering Minor : Industrial Engineering Thesis : Luminescent solar concentrators using high-contrast gratings

PROFESSIONAL EXPERIENCE

June 2025 Present	National Institute of Standards and Technology Polymers Processing Group ➤ Develop AI-agents for sustainable and novel biopharmaceutical formulations using the Autonomous Formulation Laboratory.
June 2021 May 2025	University of Washington, Seattle, WA Data Science Postdoctoral Fellow, Department of Chemical Engineering ➤ Developed closed-loop material acceleration platforms emphasizing learning faithful data representations of spectroscopy, and scattering characterizations of polymer, colloid, and soft-matter systems.
June 2017 June 2021	University at Buffalo, Buffalo, NY, USA Graduate Research Assistant ➤ Developed physics-based models for cyclic voltammetry, phase diagrams of polymer blends, and data-driven models for structure-property relations for catalysis and organic photovoltaics.
June 2016 June 2017	Industrial Internships ➤ Caterpillar India Pvt.Ltd : Finite element-based modeling of thermal stresses in automotive engine components (gaskets) ➤ Continental Automotive Components India Pvt.Ltd : Finite element-based dynamic (vibration and failure) analysis of automotive engine components
June 2016 June 2017	Indian Institute of Technology Madras, Chennai, TN, India Undergraduate Research ➤ Construction of constitutive models for diffusion-induced deformation of photopolymer under selective irradiation

TEACHING EXPERIENCE

January 2023 March 2023	Primary Instructor, Special Topics in Chemical Engineering : Design of Experiments CHEM E 599 B, Winter quarter, Evaluation 4.2/5 ➤ Designed and delivered a graduate-level course on the design of experiments with applications in materials science and chemical engineering.
March 2022 June 2022	CAPSTONE DIRECT project CHEME / CHEM / MSE 547 : Molecular Data Science Capstone ➤ Mentored five graduate students from the Department of Materials Science toward their final project titled "The infinite space of spectroscopy measurements" at the University of Washington.

PUBLICATIONS & PRE-PRINTS

Total 14 publications, 96 citations, h-index 5, i10-index 3, ORCID : 0000-0003-2998-769X

Vaddi, K., Chiang, H. T., Pozzo, L. D. “Autonomous Phase Mapping of Gold Nanoparticles Synthesis with Differentiable Models of Spectral Shape”

ChemRxiv, 2025, DOI : 10.26434/chemrxiv-2025-zktwx (**Corresponding Author**)

HT Chiang, Z Zhang, **K Vaddi**, A Tezcan, L Pozzo “Efficient Analysis of Small-Angle Scattering Curves for Large Biomolecular Assemblies Using Monte Carlo Methods”

ChemRxiv, 2025, DOI : 10.26434/chemrxiv-2025-68833

K Li, **K Vaddi**, S Seifert, J Mata, LD Pozzo “Self-assembly of a Triblock Copolymer in the Presence of a Rigid Conjugated Polyelectrolyte”

Macromolecules 57.24 (2024) : 11717-11726.

Chiang, Huat Thart, **Kiran Vaddi**, and Lilo D. Pozzo. “Data-Driven Exploration of Silver Nanoplate Formation in Multidimensional Chemical Design Spaces”

Digital Discovery, 2024, 3 (11), 2252-2264

Vaddi, Kiran, Karen Li, and Lilo D. Pozzo. “Metric geometry tools for automatic structure phase map generation.”

Digital Discovery, 2023, 2 (5), 1471-1483 (**Corresponding Author**)

Politi, M., Baum, F., **Vaddi, K.**, Vasquez, J., Antonio, E., Bishop, B. P., ..., and Pozzo, L. D. “High-Throughput Workflow for the Synthesis of CdSe Nanocrystals Using a Sonochemical Materials Acceleration Platform.”

Digital Discovery, 2023, 2, 1042-1057 (**Editor's Choice 2023**)

Vaddi, K., Liu, H., Pokuri, B. S. S., Ganapathysubramanian, B., and Wodo, O. (2023) “Construction and high-throughput exploration of phase diagrams of multi-component organic blends”

Computational Materials Science 216 (2023) : 111829.

Vaddi, Kiran, Huat Thart Chiang, Lilo D. Pozzo “Autonomous retrosynthesis of gold nanoparticles via spectral shape matching”

Digital Discovery 1.4 (2022) : 502-510 (**Editor's Choice 2022, Corresponding Author**)

Lachowski, K. J., **Vaddi, K.**, Naser, N. Y., Baneyx, F., Pozzo, L. D. “Multivariate Analysis of Peptide-Driven Nucleation and Growth of Au Nanoparticles”

Digital Discovery 1.4 (2022) : 427-439

Vaddi, Kiran, Olga Wodo “Active knowledge extraction from cyclic voltammetry”

Energies 15.13 (2022) : 4575

Elikkottil, A., **Vaddi, K.**, Reddy, K. S., Pesala, B. “Reduction of Escape Cone Losses in Luminescent Solar Concentrators Using High-Contrast Gratings.”

In Advances in Energy Research, Vol. 1 (pp. 37-43). Springer, Singapore, 2020

Vaddi, Kiran, and Olga Wodo. “Metric Learning for High-Throughput Combinatorial Data Sets.”

ACS Combinatorial Science 21.11 (2019) : 726-735 (**Corresponding Author**)

GRANT PROPOSALS

August 2025 July 2027	Theory-integrated Autonomous Experimentation for Sustainable Biotherapeutic Formulation Design U.S. Department of Commerce, NIST Materials Measurement Laboratory (Awarded FAIN-70NANB25H090-0) PI on the grant to work with NIST Autonomous Formulation Lab and develop physics-informed AI-driven agents for desinging biotherapeutic drug formulation by optimizing measurement campaigns.
June 2022 June 2025	AI-Accelerated Optimization of Self-Assembled Organic Mixed Ionic-Electronic Conductors (OMIEC) DOE- Basic Energy Sciences-Neutron Scattering (Awarded DE-SC0019911) Contributed to this PI-led Energy grant on building AI-driven workflows understanding a novel class of OMIEC materials based on conjugated polymer blends from the Department of Energy (30%).

June 2023	Accelerating the structure-based material retrosynthesis using shape representations 2022 Acceleration Consortium Project Theme : Machine learning algorithms for synthesis protocols > Led a proposal to advance self-driving laboratories for soft-matter systems as part of the University of Toronto's Acceleration Consortium, which received an honorable mention (80%).
-----------	---

AWARDS

February 2025	Future Investigator Travel Award American Physical Society Division of Soft Materials
September 2024	Outstanding Reviewer Award Royal Society of Chemistry Digital Discovery Journal
December 2023	Best Poster Award Materials Research Society Fall Meeting-Symposium on Data Science
March 2023	Data Science Conference Travel Award eScience Institute, University of Washington, Seattle, WA
June 2021	UW Data Science Postdoctoral Fellowship eScience Institute, University of Washington, Seattle, WA
August 2013	Outstanding Undergraduate Service Award National Service Scheme, Indian Institute of Technology Madras

INVITED TALKS

December 2023 Invited Panelist	The Kavli Foundation Frontiers of Materials, BOSTON, MA, USA Symposium X - How to Build a Self-Driving Lab Keith A. Brown, John Dunlap, Robert Epps, Jason Hattrick-Simpers, Kiran Vaddi
December 2023 Invited talk	Materials Research Society, BOSTON, MA, USA A Tutorial on Functional Data Analysis for High-Throughput Experiments Kiran Vaddi
October 2023 Invited talk	UW Data Science Seminar, SEATTLE, WA, USA Functional data analysis tools for autonomous experimentation Kiran Vaddi
August 2023 Invited talk	American Chemical Society Fall Meeting, SAN FRANCISCO, CA, USA Cold, warm, warmer, hot! Impact of distance metrics on autonomous experimentation Kiran Vaddi , Kacper Lachowski, Huat T Chiang, Karen Li, Lilo D. Pozzo
January 2020 Invited talk	IEEE Rochester Section Chapter, ROCHESTER, NY, USA Function Space Data Representation of Temporal Signals for Machine Learning Kiran Vaddi

SELECT CONFERENCE PRESENTATIONS & POSTERS

August 2024 Contributed talk	American Chemical Society Fall Meeting, DENVER, CO, USA Understanding Design Rules of Colloidal Self-Assembly Using Autonomous Phase Mapping Kiran Vaddi , Huat T Chiang, Lilo D. Pozzo
April 2024 Contributed talk	Materials Research Society Spring Meeting, SEATTLE, WA, USA High-Throughput Structural Investigation of Block Copolymer and Conjugated Polymer Co-Assemblies Kiran Vaddi , Karen Li, Lilo D. Pozzo
December 2023 Poster	Materials Research Society Fall Meeting, BOSTON, MA, USA Function Space Representations for Complex Material Workflows Kiran Vaddi (Best poster award)

March 2023 Contributed talk	American Physics Society March Meeting, LAS VEGAS, NV, USA Metric geometry tools for automatic structure phase map generation Kiran Vaddi, Karen Li, Lilo D. Pozzo
August 2022 Poster	Gordon Research Conference on Computational Materials Sciences, NEWRY, ME, USA Shape-based data representations for experimental spectra Kiran Vaddi, Huat T Chiang, Karen Li, Lilo D. Pozzo
May 2022 Contributed talk	Materials Research Society Spring Meeting, HONOLULU, HAWAII, USA Autonomous retrosynthesis of nanoscale structures via spectral shape matching Kiran Vaddi, Huat Thart Chiang, Lilo D. Pozzo
December 2020 Contributed talk	Materials Research Society Fall Meeting, BOSTON, MA, USA High-throughput exploration of materials-phase diagram maps in multi-component organic blends Kiran Vaddi, Balaji Pokuri, Baskar Ganapathysubramanian, Olga Wodo
October 2020 Poster	AI for Materials : From Discovery to Production, WEBINAR, Probabilistic representation of cyclic voltammetry curves for data-driven material discovery Kiran Vaddi, Olga Wodo
December 2019 Contributed talk	Materials Research Society Fall Meeting, BOSTON, MA, USA Accelerating catalyst discovery using Gaussian processes and active learning Kiran Vaddi, Olga Wodo, Krishna Rajan
May 2019 Poster	Toyota Research Institute Accelerated Material Design and Discovery Meeting, BOSTON, MA, USA Machine Learning-Based Simulation Tools for Combinatorial Experiments Kiran Vaddi, Olga Wodo, Krishna Rajan

⇒ RESEARCH MENTORING

June 2021 Present	Graduate student mentoring, Pozzo Research Group, UNIVERSITY OF WASHINGTON, Seattle, WA > Kacper Lachowski, Ph.D. > Huat That Chiang, Ph.D. candidate > Karen Li, Ph.D. candidate
September 2024 Present	Undergraduate student mentoring, Pozzo Research Group, UNIVERSITY OF WASHINGTON, Seattle, WA > Aleks Grey, Chemical Engineering.
December 2019	Partnerships for Research and Education in Materials (PREM), MRS, 2019, Boston, USA > Mentored two undergraduate material science students during their visit to the MRS 2019 Fall meeting. > Advised mentees towards successful abstract writing, poster competitions, and networking sessions.

⇒ SERVICE, OUTREACH & LEADERSHIP

June 2021 Present	Academic service > Peer review : RSC Digital Discovery (Outstanding Reviewer 2023), Neural Information Processing Systems (NeurIPS), Journal of Electroanalytical Chemistry, Matter, Nature communications, RSC Advances. > Committees : UW ChemE Distinguished Young Research Scholar Seminar reviewer (2022, 2024), Post-doc representative of 2023 faculty hiring committee.
June 2021 December 2022	Co-chair eScience Postdoc Seminar, UNIVERSITY OF WASHINGTON, Seattle, WA > Organized a year-long seminar series for the eScience Institute at the University of Washington, Seattle including invited talks about research and career development.
2020	OPEN SOURCE SOFTWARE github.com/kiranvad > pyMECSim : Simulating voltammograms of complex multi-step electrochemical reactions in Python. > geomstats : Performing geometric data analysis on functional data. > polyphase : Computing phase diagrams of polymer blends using Flory-Huggins theory. > Computational Topology : Example notebooks introducing topological data analysis.

- June 2012 | **National Social Service Scheme, IIT MADRAS, Chennai, India**
- June 2015
- > Awarded **Outstanding Undergraduate Service Award** in 2013 for organizing and mentoring the 'IIT Madras for The Banyan' project.
 - > Co-organized the inaugural, student-led, interactive support sessions for persons living through poverty and homelessness via [The Banyan](#)
 - > Developed teaching materials and actively participated in community teaching programs.



PROFESSIONAL MEMBERSHIPS

- 2018 Materials Research Society
- 2023 Americal Physical Society (DPOLY, DSOF, GDS)
- 2023 American Institute of Chemical Engineers (CoMSEF, MESD)
- 2023 American Chemical Society (COMP, PMSE)